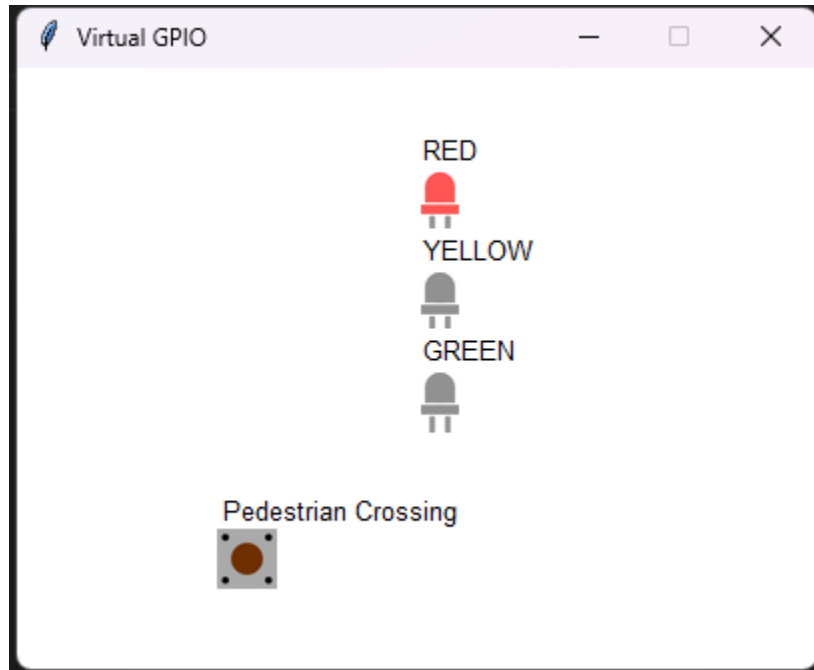




traffic_light_sim.py

```
from tkgpio import TkCircuit

configuration = {
    "width": 400,
    "height": 300,
    "leds": [
        {"x": 200, "y": 50, "name": "RED", "pin": 17},
        {"x": 200, "y": 100, "name": "YELLOW", "pin": 27},
        {"x": 200, "y": 150, "name": "GREEN", "pin": 22}
    ],
    "buttons": [
        {"x": 100, "y": 230, "name": "Pedestrian Crossing", "pin": 11}
    ],
    "title": "Traffic Light Simulation"
}

circuit = TkCircuit(configuration)

@circuit.run
def main():
    import traffic_light

if __name__ == "__main__":
    print("Starting Traffic Light Simulation...")
```

traffic_light_sim.py

```

from gpiozero import LED, Button
from time import sleep

red_light = LED(17)
yellow_light = LED(27)
green_light = LED(22)

pedestrian_button = Button(11)

# initial
red_light.on()
yellow_light.off()
green_light.off()

auto_mode = True
red_time = 5
yellow_time = 2
green_time = 5
pedestrian_requested = False

def pedestrian_crossing_request():
    global pedestrian_requested
    print("Pedestrian requested to cross!")
    pedestrian_requested = True

# กดปุ่ม
pedestrian_button.when_pressed = pedestrian_crossing_request

def traffic_light_sequence():
    global pedestrian_requested

    while True:
        # ไฟแดง
        red_light.on()
        yellow_light.off()
        green_light.off()
        print("Red light: Stop")
        sleep(red_time)

        # ไม่กด
        if auto_mode or pedestrian_requested:
            # ไฟเขียว
            red_light.off()
            yellow_light.off()
            green_light.on()
            print("Green light: Go")
            sleep(green_time)

```

```
# ไฟเหลือง
red_light.off()
yellow_light.on()
green_light.off()
print("Yellow light: Prepare to stop")
sleep(yellow_time)

pedestrian_requested = False

try:
    traffic_light_sequence()
except KeyboardInterrupt:
    red_light.off()
    yellow_light.off()
    green_light.off()
    print("\nTraffic light simulation stopped")
```